Titanic Survival Prediction: Project Report

A machine learning study of which passengers survived the 1912 Titanic shipwreck, based on passenger records (name, age, sex, class, fare, family, cabin, port of embarkation).

1. Summary

This project builds an end-to-end machine learning pipeline and interactive web application to predict whether a Titanic passenger survived the 1912 disaster, and uses statistical modeling to describe what characteristics governed survival. The production model is a tuned, balanced **Logistic Regression pipeline** with automated feature engineering, imputations, standard scaling, and categorical one-hot encoding, achieving **83.2% validation accuracy** on a stratified hold-out test set.

The main findings demonstrate that survival strictly followed the traditional maritime doctrine of *"women and children first"*, heavily structured by socioeconomic status:

- **Women and young boys** (carrying the honorific title *Master*) in 1st and 2nd class had survival rates between **92% and 100%**.
- **Adult men** in 2nd and 3rd class rarely survived (8.8% and 11.3% survival rates respectively).
- **Class privilege** was decisive across every group: 1st-class passengers were dramatically more likely to secure lifeboat accommodations due to proximity to the upper boat deck.

2. Data

The project utilizes the canonical passenger dataset (`train.csv`), containing historical records for 891 individuals.

File	Rows	Use
`train.csv`	891	Labelled passengers (`Survived = 0 or 1`), used for feature extraction, training, and hold-out validation.

Of the 891 labelled passengers, **38.4% survived (342 passengers)**, presenting a moderately imbalanced binary classification target.

Missing values in the training set

Column	Missing	Handling
Age	177 (19.9%)	Imputed using the median age across all passengers via `SimpleImputer(strategy="median")`.
Cabin	687 (77.1%)	Excluded due to severe sparsity (>77% missing).
Embarked	2 (0.2%)	Imputed with the most frequent embarkation port (Southampton, `'S'`) via `SimpleImputer(strategy="most_frequent")`.

3. Method

3.1 Feature engineering

Beyond raw tabular columns, several domain-specific features were engineered to capture family dynamics and social hierarchy:

- **Title extracted from name:** Extracted honorific titles (*Mr*, *Mrs*, *Miss*, *Master*, *Dr*, *Rev*, etc.) from each passenger's full name. Title simultaneously captures gender, social status, age group, and marital status in a single field.
- **FamilySize and IsAlone:** Calculated family size as `SibSp + Parch + 1` (siblings, spouses, parents, children, plus self), accompanied by a boolean flag `IsAlone` indicating whether the passenger travelled unaccompanied.
- **Numerical transformations:** Standardized numerical attributes (`Pclass` , `Age` , `Fare` , `FamilySize` , `IsAlone`) using `StandardScaler` after median imputation.
- **Categorical one-hot encoding:** Encoded nominal variables (`Sex` , `Embarked` , `Title`) into binary indicators via `OneHotEncoder(handle_unknown="ignore")`.

3.2 Model pipeline & training

The full workflow was constructed into an atomic, leak-free Scikit-Learn `Pipeline` consisting of a `ColumnTransformer` and a regularized `LogisticRegression` classifier.

To mitigate class imbalance and ensure high recall for survivors, the classifier was configured with `class_weight="balanced"` and an increased iteration threshold (`max_iter=2000`). The dataset was evaluated using a **stratified 80/20 train-test split** (712 train, 179 validation) with fixed random seed (`random_state=42`).

4. Results

4.1 Final model evaluation

Measure	Score
Hold-out validation accuracy (20% of data)	**83.2%** (149 / 179 correct)
Macro average F1-score	**0.82**
Weighted average F1-score	**0.83**

Hold-out detail

Class	Precision	Recall	F1-Score	Passengers (Support)
Did not survive (0)	0.88	0.85	0.86	110
Survived (1)	0.77	0.81	0.79	69

Confusion Matrix:

```
                    Predicted Died    Predicted Survived
Actual Died (110)           93                  17
Actual Survived (69)        13                  56
```

By utilizing balanced class weights, the model achieved an **81% recall on survivors** (56 out of 69 detected) while maintaining high precision on casualties (**88%**), preventing dangerous false-confidence bias.

5. Who survived?

5.1 What drives the predictions

The model's learned regression coefficients indicate how strongly each feature shifts the log-odds of passenger survival:

Feature / Variable	Coefficient	Effect	Interpretation
`Title_Master`	**+1.96**	Strong Positive	Young male boys were prioritized for lifeboats along with women.
`Sex_female`	**+1.06**	Strong Positive	Strict enforcement of "women and children first".
`Title_Mrs`	**+0.58**	Positive	Married adult women were consistently given evacuation priority.
`Fare`	**+0.19**	Positive	Wealthier ticket holders had cabins with faster lifeboat deck access.
`Age`	**-0.33**	Negative	Older passengers faced reduced odds compared to younger travelers.
`Pclass`	**-0.77**	Negative	Higher class number (e.g. 3rd class) substantially lowered survival odds.
`FamilySize`	**-0.81**	Negative	Large families faced severe logistical hurdles gathering at muster stations.
`Title_Rev`	**-0.97**	Strong Negative	Ministers and clergy notably remained behind assisting others.
`Sex_male` / `Title_Mr`	**-0.99** / **-0.83**	Strong Negative	Adult men faced the steepest evacuation barrier across the entire ship.

5.2 Observed survival rates

These are empirical survival rates directly calculated from the training data:

Group	1st class	2nd class	3rd class	Overall
Women	97%	92%	50%	74%
Boys (title "Master")	100%	100%	39%	58%
Men (title "Mr")	35%	8%	11%	16%
All Adult Men	37%	16%	14%	19%

Two critical patterns emerge:

- **Class mattered intensely for women and children:** A woman in 1st class had nearly double the survival rate of a woman in 3rd class (97% vs 50%). Similarly, 100% of 1st and 2nd class young boys survived, compared to only 39% in steerage.
- **1st-class men outperformed other men:** 1st-class men survived at triple to quadruple the rate of 2nd and 3rd-class men (35% vs 8%–11%), directly reflecting cabin deck proximity and

preferential early access to lifeboats.

6. Limitations

- **Small dataset:** With 891 labelled passengers, metric estimates can vary by ±1.5 percentage points across different cross-validation folds.
- **Ceiling on tabular accuracy:** Accuracies around 82%–84% represent the practical limit for this dataset. Key determinants—such as physical corridor clearance, locked steerage gates, lifeboat lowering times, and individual human choices—are absent from passenger manifest records.
- **Imputed ages:** 177 passengers (20%) lacked verified age entries and received median imputation, introducing variance for teens and senior citizens.
- **Correlation, not causation:** The model highlights statistical regularities in passenger survival, but does not simulate physical evacuation mechanics.

7. Deliverables

File	Description
`app.py`	Interactive Streamlit web application providing real-time inference and survival probability scoring.
`train_model.py`	Automated ML pipeline script: preprocessing, pipeline creation, hold-out evaluation, and model serialization.
`titanic_survival_model.joblib`	Serialized production Scikit-Learn pipeline (feature transformers + trained classifier).
`evaluation_report.txt`	Validation accuracy, classification report, and confusion matrix summary.
`metrics.json`	Structured JSON containing validation accuracy and 2×2 confusion matrix array.
`train.csv`	The 891-row training dataset containing labelled passenger records.
`requirements.txt`	Pinned dependencies (`pandas` , `scikit-learn` , `joblib` , `streamlit`).
`PROJECT_REPORT.md`	Comprehensive project report in Markdown format.
`PROJECT_REPORT.html`	Print-ready, styled HTML report (compatible with Print to PDF).

8. Conclusion

Gender, social class, and family structure were the decisive determinants of survival during the 1912 Titanic disaster. By synthesizing automated feature engineering with a balanced Logistic Regression pipeline, this project delivers an accurate (**83.2% accuracy**), robust, and transparent machine learning solution.

Paired with the interactive **Streamlit** application, the project delivers both high academic rigor and intuitive practical software, enabling accessible exploration of historical demographic insights.